



Department of Computer Science

Namal University, Mianwali

“Final Year Project Management System”

Database Design Document

Course: Database System

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REVISION HISTORY

Date	Version	Description
15/05/26	V 1.0	Initial submission of Database Design Document for FYP Management System. Includes complete data dictionary with data types and constraints, business rules, Entity Relationship Diagram (ERD) with all relationships and cardinalities, multi-value attribute handling through separate tables associative entities for many-to-many relationships, sample data for testing, and test cases validating complete FYP lifecycle from student registration to evaluation results.

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CHAPTER 1: PROJECT OVERVIEW

1.1. INTRODUCTION:

The Final Year Project (FYP) is an extremely significant and mandatory component of the undergraduate degree program at Namal University. It allows students to combine practical and theoretical knowledge to solve real-world problems. It is the final major requirement before degree completion, making it important for both students and the university.

Managing the FYP process is not easy at any university. Currently, all FYP-related work is done manually at Namal University. Students submit their project ideas on paper and wait for coordinator approval. There is no proper follow-up system to track which group is at which stage. The university stores FYP data in Excel sheets and other manual tools to track the entire process from group formation to final evaluation.

This manual method causes several issues. Data is repeated across multiple files, making it inconsistent and difficult to manage. Records are often lost or misplaced. The entire process is slow and ineffective. As the number of students increases each year, this manual approach becomes harder to handle.

The purpose of the FYP Management System is to digitize and automate all FYP-related activities at Namal University. The system will provide a centralized platform where students can submit their proposals online, coordinators can approve them, and supervisors can monitor progress. The system will track each group from formation to final defense, ensuring no stage is missed.

The system will automate key processes including student group formation, project proposal submission, supervisor and co-supervisor assignment, milestone creation, submission uploads, feedback provision, evaluation scheduling, and result recording. It will also maintain complete records of all past and current projects for future reference.

The system is specifically developed for Namal University, Mianwali, Pakistan. Namal University is a higher education institution that offers undergraduate degree programs in computer science and related fields. The university has a growing number of students each year, making manual FYP management increasingly difficult.

The system will operate in a university academic environment. It will be accessed by students, faculty members, department coordinators, and external evaluators from organizations.

1.2. PROBLEM STATEMENT:

All tasks associated with Final Year Projects at Namal University are currently handled using manual processes or basic Excel-based programs. The management team, including coordinators and faculty, struggles with numerous challenges due to the rising number of students and projects each year.

Issues:

1. There is no centralized platform to manage the FYP process from student group formation to final defense.
2. Data redundancy occurs as the same information is stored in multiple files and spreadsheets.
3. Data inconsistency exists because different versions of the same data are maintained separately.
4. Loss of records is common as manual documents can be easily misplaced or destroyed.
5. Significant administrative overhead is placed on coordinators and faculty members.
6. Time-consuming processes delay approvals and feedback.
7. Limited scalability means the current system cannot handle increasing number of students each year.
8. Absence of proper record management makes it hard to retrieve past project information.

Impact on Users:

1. Experience delayed proposal approvals, lack of submission transparency, and untimely feedback, hindering their academic progress.
2. Bear excessive administrative burden, reducing time available for meaningful student supervision and quality assurance.
3. Face difficulty accessing submission records, scheduling evaluations, and generating reliable performance reports.

1.3. PROJECT OBJECTIVES:

General Objective:

To design and implement a fully functional, web-based FYP Management System with a centralized relational database that automates and streamlines the entire FYP lifecycle at Namal University, ensuring efficiency, data integrity, and role-based accessibility for all stakeholders.

Specific Objectives:

1. To computerize the entire FYP management process, from student group formation to final assessment, eliminating manual paperwork and reducing processing time by at least 50%.
2. To grant role-specific access to students, supervisors, coordinators, and evaluators, ensuring each user can only view and perform actions relevant to their designated role.
3. To maintain comprehensive and permanent records of all past and present FYPs in a single centralized database, enabling quick retrieval and historical analysis.
4. To design and implement a structured relational database follow normalization standards for efficient storage, retrieval and management of all FYP-related data.

5. To remove all forms of data duplication by storing each piece of information only once, ensuring uniformity and consistency across the entire system.
6. To build a scalable system architecture capable of supporting increasing numbers of users, groups, and projects without degradation in performance or response time.
7. To maintain data integrity, consistency, and security through the use of primary and foreign key constraints, normalization, validation rules, and authentication-based access control mechanisms.

1.4. GITHUB REPOSITORY LINK:

LINK: <https://github.com/mubashir-hassan-git/FYP-Management-System>

CHAPTER 2: DETAILED DATABASE DESIGN

2.1. ENTITY:

The following entities are central to the FYP Management System database. Each entity represents a real-world object, concept, or event that the system tracks.

Sr. No	Entity Name	Description
01	PERSON	A person is an individual who interacts with the system. Persons can be Students, Faculty members or External Evaluators. The PERSON entity serves as the supertype for all user types.
02	STUDENT	A student is a person enrolled in a degree program who is required to complete a Final Year Project. Students submit project work, and receive evaluations.
03	FACULTY	A faculty member is a person employed by the university who serves as a supervisor or co-supervisor for student projects.
04	COORDINATOR	A coordinator is a faculty member appointed to manage FYP activities for a specific department. Coordinators create groups, assign projects, assign supervisors, and create milestones.
05	EXTERNAL_EVALUATOR	An external evaluator is a person from industry or another institution who evaluates student projects based on their domain expertise.
06	EVALUATORS	An evaluator is a person (internal faculty or external) who is assigned to evaluate student projects. This entity unifies internal and external evaluators.
07	DEPARTMENT	A department is an academic unit within the university that offers degree programs and manages FYPs.
08	BATCH	A batch represents a group of students admitted in a specific academic session .
09	GROUP	A group consists of two students who work together on a single project. Each group is assigned one project and has supervisor and co-supervisor.
10	PROJECT	A project is a specific FYP undertaken by a group. It has a title, description, domain, and status.
11	MILESTONE	A milestone represents a key deliverable in the FYP lifecycle. Milestones are defined per batch.
12	SUBMISSION	A submission is the work uploaded by a student group for a specific milestone.
13	FEEDBACK	Feedback is the review provided by a supervisor or co-supervisor on a student submission.

14	EVALUATION	An evaluation is an assessment event conducted for a specific milestone and group.
15	EVALUATION_TYPE	An evaluation type defines the category of evaluation with its weight age.
16	EVALUATION_CRITERIA	Evaluation criteria are the rubrics used to assess a project. Each criterion has a name and maximum marks.
17	EVALUATION_RESULT	An evaluation result records the marks obtained by a group for each criterion, along with remarks from evaluators.
18	EVALUATION_ASSIGNMENT	An evaluation assignment links evaluators to evaluations, specifying their role.
19	USER_ACCOUNT	A user account provides login credentials for a person to access the system.
20	ROLE	A role defines the permissions and access level of a user in the system.

2.2. DATA DICTIONARY:

2.2.1. PERSON

Sr. No	Name	Data Type	Constraint	Description
01	PersonID	INT	PRIMARY KEY, NOT NULL,	Unique identifier for each person
02	FirstName	VARCHAR	NOT NULL	First name of the person
03	LastName	VARCHAR	NULL	Last name of the person
04	Email	VARCHAR	NOT NULL, UNIQUE	Email address of the person
05	CNIC	VARCHAR	NOT NULL, UNIQUE	National ID card number
06	Gender	ENUM	NULL	Gender of the person
07	DOB	DATE	NULL	Date of birth
08	Person Type	ENUM	NOT NULL	Discriminator for subtype

2.2.2. PERSON_PHONE

Sr. No	Name	Data Type	Constraint	Description
01	PhoneID	INT	PRIMARY KEY, NOT NULL	Unique identifier for phone record
02	PersonID	INT	NOT NULL, FOREIGN KEY	References the person owning this phone
03	PhoneNumber	VARCHAR	NOT NULL	Phone number

04	PhoneType	ENUM	NOT NULL, DEFAULT 'Mobile'	Type of phone number
05	IsPrimary	BOOLEAN	NOT NULL, DEFAULT FALSE	Indicates if this is the primary phone

2.2.3. PERSON_ADDRESS

Sr. No	Name	Data Type	Constraint	Description
01	AddressID	INT	PRIMARY KEY, NOT NULL	Unique identifier for address record
02	PersonID	INT	NOT NULL, FOREIGN KEY	References the person owning this address
03	AddressLine1	VARCHAR	NOT NULL	First line of address
04	AddressLine2	VARCHAR	NULL	Second line of address
05	City	VARCHAR	NOT NULL	City name
06	Province	VARCHAR	NULL	Province or state name
07	PostalCode	VARCHAR	NULL	Postal code
08	Country	VARCHAR	NOT NULL, DEFAULT 'Pakistan'	Country name
09	IsPrimary	BOOLEAN	NOT NULL, DEFAULT FALSE	Indicates if this is the primary address

2.2.4. STUDENT

Sr. No	Name	Data Type	Constraint	Description
01	Registration_No	VARCHAR	NOT NULL, UNIQUE	University registration number
02	BatchID	INT	NOT NULL, FOREIGN KEY	References the batch
03	GroupID	INT	NULL, FOREIGN KEY	References the assigned group

2.2.5. FACULTY

Sr. No	Name	Data Type	Constraint	Description
03	EmployeeNo	VARCHAR	NOT NULL, UNIQUE	Staff ID number

04	Designation	VARCHAR	NULL	Job title (Professor, Lecturer, etc.)
05	DepartmentID	INT	NOT NULL, FOREIGN KEY	References the department

2.2.6. COORDINATOR

Sr. No	Name	Data Type	Constraint	Description
01	CoordinatorID	INT	PRIMARY KEY, NOT NULL	Unique identifier for coordinator
02	DepartmentID	INT	NOT NULL, FOREIGN KEY	References the department managed
03	Appointment_Date	DATE	NOT NULL	Date when coordinator was appointed

2.2.7. EXTERNAL_EVALUATOR

Sr. No	Name	Data Type	Constraint	Description
01	Organization	VARCHAR	NOT NULL	Company or organization name
02	Designation	VARCHAR	NULL	Job title at the organization

2.2.8. DEPARTMENT

Sr. No	Name	Data Type	Constraint	Description
01	DepartmentID	INT	PRIMARY KEY, NOT NULL	Unique identifier for department
02	Department_Name	VARCHAR	NOT NULL, UNIQUE	Name of the department

2.2.9. BATCH

Sr. No	Name	Data Type	Constraint	Description
01	BatchID	INT	PRIMARY KEY, NOT NULL	Unique identifier for batch
02	BatchName	VARCHAR	NOT NULL, UNIQUE	Name of the batch
03	StartYear	YEAR	NOT NULL	Starting year of the batch
04	EndYear	YEAR	NOT NULL	Expected graduation year

05	Session	ENUM	NOT NULL	Academic session
06	DepartmentID	INT	NOT NULL, FOREIGN KEY	References the department

2.2.10. GROUPS

Sr. No	Name	Data Type	Constraint	Description
01	GroupID	INT	PRIMARY KEY, NOT NULL	Unique identifier for group
02	GroupName	VARCHAR	NOT NULL, UNIQUE	Name of the group
03	CoordinatorID	INT	NOT NULL, FOREIGN KEY	References the coordinator
04	ProjectID	INT	NULL, FOREIGN KEY	References the assigned project

2.2.11. PROJECT

S	Name	Data Type	Constraint	Description
01	ProjectID	INT	PRIMARY KEY, NOT NULL	Unique identifier for project
02	Title	VARCHAR	NOT NULL	Title of the project
03	Description	VARCHAR	NULL	Detailed description of the project
04	Domain	VARCHAR	NULL	Project domain
05	ProjectStatus	ENUM	NOT NULL, DEFAULT 'Proposed'	Current status

2.2.12. PROJECT_SUPERVISION

Sr. No	Name	Data Type	Constraint	Description
01	ProjectID	INT	PRIMARY KEY, FOREIGN	References the project

			KEY	
02	FacultyID	INT	PRIMARY KEY, FOREIGN KEY	References the faculty member
03	Role	ENUM	NOT NULL	Role of the faculty

2.2.13. MILESTONE

Sr. No	Name	Data Type	Constraint	Description
01	MilestoneID	INT	PRIMARY KEY, NOT NULL	Unique identifier for milestone
02	BatchID	INT	NOT NULL, FOREIGN KEY	References the batch
03	Title	VARCHAR	NOT NULL	Title of the milestone
04	Description	VARCHAR	NULL	Detailed description
05	DueDate	DATE	NOT NULL	Submission deadline
06	Weightage	DECIMAL(5,2)	NOT NULL, DEFAULT 0	Percentage weightage
07	Status	ENUM	NOT NULL, DEFAULT 'Active'	Current status

2.2.14. SUBMISSION

Sr. No	Name	Data Type	Constraint	Description
01	SubmissionID	INT	PRIMARY KEY, NOT NULL	Unique identifier for submission
02	ProjectID	INT	NOT NULL, FOREIGN KEY	References the project
03	MilestoneID	INT	NOT NULL, FOREIGN KEY	References the milestone
04	SubmissionVersion	VARCHAR(10)	NOT NULL, DEFAULT 'v1.0'	Version number
05	SubmissionType	ENUM	NOT NULL	Type of submission
06	Status	ENUM	NOT NULL	Tell the status of Submissions

2.2.15. SUBMISSION_FILE

Sr. No	Name	Data Type	Constraint	Description
01	FileID	INT	PRIMARY KEY, NOT NULL	Unique identifier for file
02	SubmissionID	INT	NOT NULL, FOREIGN KEY	References the submission
03	FileName	VARCHAR	NOT NULL	Original name of the file
04	FilePath	VARCHAR	NOT NULL	Storage path of the file
05	FileType	VARCHAR	NULL	MIME type
06	FileSize	INT	NULL	Size of file

2.2.16. FEEDBACK

Sr. No	Name	Data Type	Constraint	Description
01	FeedbackID	INT	PRIMARY KEY, NOT NULL	Unique identifier for feedback
02	SubmissionID	INT	NOT NULL, FOREIGN KEY	References the submission
03	FacultyID	INT	NOT NULL, FOREIGN KEY	References faculty
04	Comment	VARCHAR	NULL	Feedback comments
05	FeedbackType	ENUM	NOT NULL	Type of feedback
06	Date	DATE	NOT NULL, DEFAULT CURRENT_DATE	Date of feedback

2.2.17. EVALUATION_TYPE

Sr. No	Name	Data Type	Constraint	Description
01	TypeID	INT	PRIMARY KEY, NOT NULL	Unique identifier for evaluation type
02	TypeName	VARCHAR	NOT NULL, UNIQUE	Name of evaluation type
03	Weightage	DECIMAL	NOT NULL	Percentage weightage

2.2.18. EVALUATION

Sr. No	Name	Data Type	Constraint	Description
01	EvaluationID	INT	PRIMARY KEY, NOT NULL	Unique identifier for evaluation
02	MilestoneID	INT	NOT NULL, FOREIGN KEY	References milestone
03	GroupID	INT	NOT NULL, FOREIGN KEY	References group
04	EvaluationDate	DATE	NOT NULL	Date of evaluation
05	Status	ENUM	NOT NULL, DEFAULT 'Scheduled'	Current status

2.2.19. EVALUATION_CRITERIA

Sr. No	Name	Data Type	Constraint	Description
01	CriteriaID	INT	PRIMARY KEY, NOT NULL	Unique identifier for criteria
02	EvaluationID	INT	NOT NULL, FOREIGN KEY	References evaluation
03	CriteriaName	VARCHAR	NOT NULL	Name of criteria
04	MaxMarks	DECIMA	NOT NULL	Maximum marks
05	Description	VARCHAR	NULL	Detailed description

2.2.20. EVALUATION_RESULT

Sr. No	Name	Data Type	Constraint	Description
01	ResultID	INT	PRIMARY KEY, NOT NULL	Unique identifier for result
02	EvaluationID	INT	NOT NULL, FOREIGN KEY	References evaluation
03	GroupID	INT	NOT NULL, FOREIGN KEY	References group
04	CriteriaID	INT	NOT NULL,	References criteria

			FOREIGN KEY	
05	ObtainedMarks	DECIMAL	NOT NULL	Marks obtained
06	Remarks	VARCHAR	NULL	Evaluator comments

2.2.21. EVALUATORS

Sr. No	Name	Data Type	Constraint	Description
01	EvaluatorID	INT	PRIMARY KEY, NOT NULL	Unique identifier for evaluator
02	PersonID	INT	NOT NULL, UNIQUE, FOREIGN KEY	References the person
03	Evaluator Type	ENUM	NOT NULL	Type of evaluator

2.2.22. EVALUATION_ASSIGNMENT

Sr. No	Name	Data Type	Constraint	Description
01	EvaluationID	INT	NOT NULL, FOREIGN KEY	References evaluation
02	EvaluatorID	INT	NOT NULL, FOREIGN KEY	References evaluator
03	EvaluatorRole	ENUM	NOT NULL	Role of evaluator
04	Assigned_Date	DATE	NOT NULL, DEFAULT CURRENT_DATE	Assignment date

2.2.23. USER_ACCOUNT

Sr. No	Name	Data Type	Constraint	Description
01	AccountID	INT	PRIMARY KEY, NOT NULL	Unique identifier for user account
02	Username	VARCHAR	NOT NULL, UNIQUE	Login username
03	Password	VARCHAR	NOT NULL	store password
04	Status	ENUM	NOT NULL, DEFAULT 'Active'	Account status
05	PersonID	INT	NOT NULL, UNIQUE, FOREIGN KEY	References the person

2.2.24. ROLE

Sr. No	Name	Data Type	Constraint	Description
01	RoleID	INT	PRIMARY KEY, NOT NULL	Unique identifier for role
02	RoleName	VARCHAR	NOT NULL, UNIQUE	Name of role
03	Description	VARCHAR	NULL	Description of role permissions

2.2.25. USERACCOUNT_ROLE

Sr. No	Name	Data Type	Constraint	Description
01	AccountID	INT	PRIMARY KEY, FOREIGN KEY	References user account
02	RoleID	INT	PRIMARY KEY, FOREIGN KEY	References the role

2.3. BUSINESS RULES:

The following business rules define how the FYP Management System operates according to the real-world policies and requirements of Namal University. These rules ensure data consistency, integrity, and correct system behavior.

User and Account Rules:

1. A person can have only one user account in the system.
2. A user account must have at least one role assigned to it.
3. A person can have multiple phone numbers and multiple addresses.
4. Only faculty members can become coordinators.
5. A coordinator manages FYP activities for exactly one department.
6. Each department has exactly one active coordinator at any given time.

Student and Group Rules:

1. A student must belong to exactly one batch and one department.
2. A student must be part of exactly one group.
3. A group must contain exactly two students.
4. A group can have only one active project at a time.
5. A group cannot be formed without coordinator approval.

6. Only a coordinator can create, approve, or disband a group.

Project and Supervisor Rules:

1. A project must be assigned to exactly one group.
2. A group can work on only one project at a time.
3. A project must have supervisor and co-supervisor.
4. A supervisor can supervise multiple projects simultaneously.
5. A faculty member can be supervisor for some projects and co-supervisor for others.
6. A project cannot be assigned to a group without an approved supervisor.
7. Only a coordinator can assign projects and supervisors to groups.

Milestone and Submission Rules:

1. Milestones are defined per batch. All groups in the same batch follow the same milestone schedule.
2. A milestone must have a due date and weight age defined before students can submit.
3. A group can submit multiple versions for the same milestone.
4. A submission must be associated with exactly one project and one milestone.
5. A submission can have multiple files attached.
6. Students can only submit work for milestones that are currently active.
7. A supervisor provide feedback on a submission.

Feedback Rules:

1. Only the assigned supervisor and co-supervisor can provide feedback on a group's submission.
2. Feedback can be provided multiple times on the same submission as students revise their work.

Evaluation Rules:

1. An evaluation is created for a specific milestone and group.
2. An evaluation must have at least one evaluation criterion defined.
3. Each evaluation criterion has a maximum marks value.
4. Multiple evaluators can be assigned to the same evaluation.
5. An evaluator can be assigned to multiple evaluations.
6. Evaluation results must be recorded for each criterion separately.
7. Students can view their evaluation results only after the evaluation status is marked as completed.

LINK:<https://online.visual-paradigm.com/w/qbaqjajq/diagrams/#diagram:workspace=qbaqjajq&proj=0&id=3&type=ERDiagram>



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APPENDIX: AI USAGE AND PROMPTS

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This section documents the use of Artificial Intelligence (AI) tools in the development of this Database Design Document. All AI tools were used as assistants to enhance productivity, improve clarity, and ensure technical accuracy. The prompts listed below represent the main interactions with each AI tool.

AI Tool Usage Record – FYP Management System

The following table documents the use of AI tools during the development and documentation of the FYP Management System database project.

- | | | |
|----|----------|---|
| 01 | ChatGPT | Generate initial document structure, “Create a database design document template for an FYP Management System with entities, data dictionary, business rules, and ERD sections following academic guidelines using provided template” |
| 02 | ChatGPT | Create data dictionary Generate a complete data dictionary for PERSON, STUDENT, FACULTY, PROJECT, SUBMISSION, and EVALUATION tables with appropriate data types, constraints, and descriptions. |
| 03 | Deepseek | Define business rules Identify and list all business rules for a university FYP Management System including rules for student groups, project supervision, milestone submissions, feedback, and evaluation processes. |
| 04 | ChatGPT | Format entities section List all entities for an FYP Management System database without including associative entities. Provide clear descriptions for each entity. |
| 05 | Claude | Improve problem statement Rewrite the problem statement to be more professional and structured. Include current issues, user impacts, and proposed solution in clear points. |
| 06 | ChatGPT | March 22, 2026 Generate sample data Create sample data rows for all 27 tables in the FYP Management System database with realistic values for testing. |
| 07 | Deepseek | “Validate relationships Verify the foreign key relationships and cardinalities in the ERD. Ensure all many-to-many relationships are resolved through associative tables” |
| 08 | ChatGPT | Create test cases “Develop test cases covering the complete FYP lifecycle from student registration to evaluation results to validate the database design” |
| 09 | Deepseek | Format references “Format the references section according to IEEE citation style for books, online sources, and GitHub repositories”. |
| 10 | ChatGPT | Improve introduction “Write a concise introduction paragraph covering problem domain, purpose, environment, key processes, and real-world relevance of the FYP Management System” |